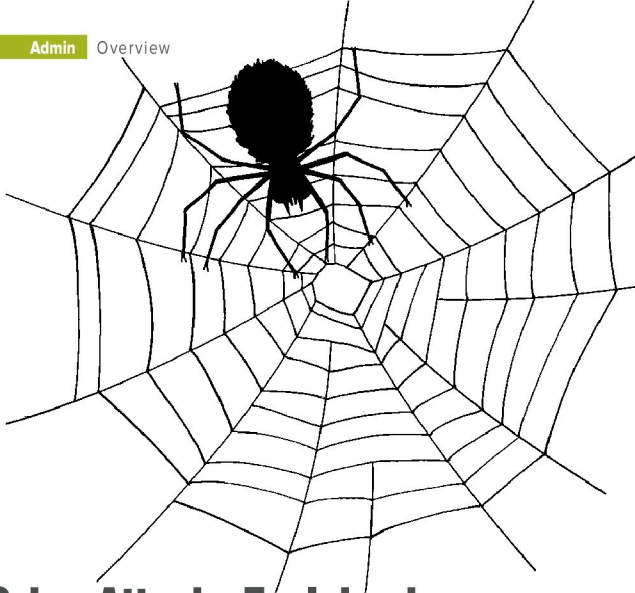




Cyber Attacks Explained

Web Exploitation

Part—5

Websites are no longer merely about having an 'Internet presence' today, but are also used for commercial transactions and to transfer sensitive data. Such wide usage helps hackers gain more knowledge on vulnerabilities and exploitation techniques than before. Various security studies show that hacking websites to gain fame or money is definitely on the rise. This article explains various Web vulnerabilities and the attacks that exploit them. We will also learn a few techniques that could be incorporated by systems administrators to protect a firm's Web infrastructure.

Before discussing how Web servers are hacked, let us first look into the various components that form a complete Web portal. To begin, a Web server is a service that typically listens on Port 80. The client software, usually a browser, connects to the port and sends HTTP queries. The Web service responds by providing the requested content, such as HTML, Javascript, etc. In

some cases, the service could be configured to run on ports other than the default one, as a small step towards security. Web servers also may host services such as FTP or NNTP, which run on their own separate default port. Figure 1 shows how Web services map into the OSI layers. The HTTP protocol works at Layer 7, whereas HTTPS (Secure Socket Layer) works at Layer 6.